

Chemical, microbial and physical evaluation of commercial bottled waters in greater Houston area of Texas.



Due to the increased demand and consumption of bottled water in the United States, there has been a growing concern about the quality of this product. Retail outlets sell local as well as imported bottled water to consumers. Three bottles for each of 35 different brands of bottled water were randomly collected from local grocery stores in the greater Houston area. Out of the 35 different brands, 16 were designated as spring water, 11 were purified and/or fortified tap water, 5 were carbonated water and 3 were distilled water. Chemical, microbial and physical properties of all samples were evaluated including pH, conductivity, bacteria counts, anion concentration, trace metal concentration, heavy metal and volatile organics concentration were determined in all samples. Inductively coupled plasma/mass spectrometry (ICPMS) was used for elemental analysis, gas chromatography with electron capture detector (GCECD) as well as gas chromatography mass spectrometry (GCMS) were used for analysis of volatile organics, ion chromatography (IC) and selective ion electrodes were used for the analysis of anions. Bacterial identification was performed using the Biolog software (Biolog, Inc., Hayward, Ca, USA). The results obtained were compared with guidelines of drinking water recommended by the International Bottled Water Association (IBWA), United States Food and Drug Administration (FDA), United States Environmental Protection Agency (EPA) and the World Health Organization (WHO) drinking water standard. The majority of the analyzed chemicals were below their respective drinking water standards for maximum admissible concentrations (MAC). Volatile organic chemicals were found to be below detection limits. Four of the 35 brands of the bottled water samples analyzed were found to be contaminated with bacteria.